

Prediction for Economists

Prediction as an objective

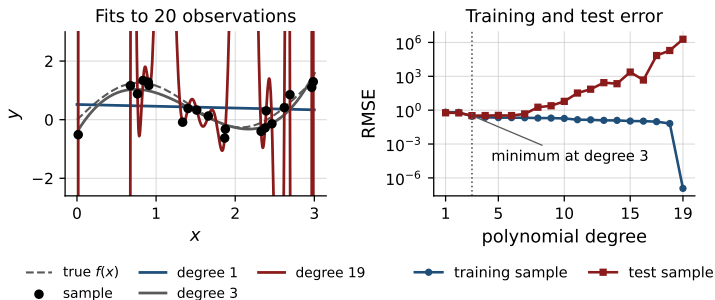
- A rule: observed characteristics in, an unobserved outcome out.
- Estimated on one sample, judged on another.
- The criterion is error on the second, not the value of any coefficient.
- Bias in the coefficients is acceptable when it lowers that error.

Everything that follows is about estimating that error honestly.

Outline

1. Overfitting and the trade-off
2. Estimating out-of-sample error
3. Leakage
4. Measures of accuracy
5. Constructing regressors
6. Choosing an estimator
7. Putting it together

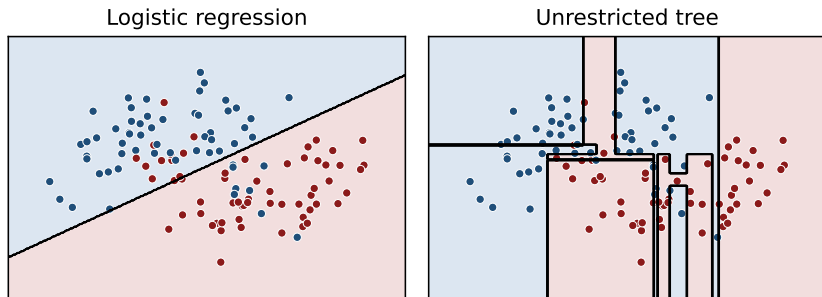
Overfitting



- In-sample fit: better at every degree.
- 20 observations, degree 19: every point exactly.
- Error in an independent sample: lowest at a low degree.

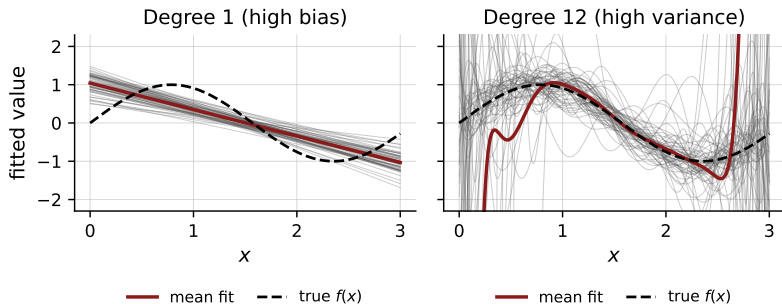
Fit in the estimation sample is not evidence of accuracy elsewhere.

Overfitting with a binary outcome



- Not a property of polynomials, nor of continuous outcomes.
- The unrestricted tree isolates single observations.
- Its boundary is this sample, not a regularity.

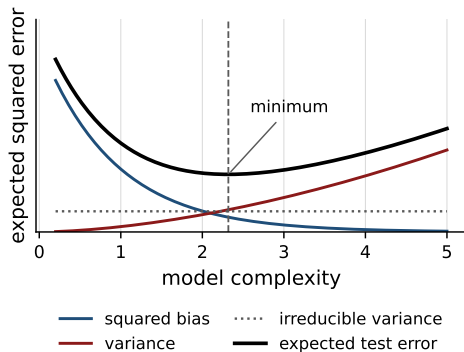
Bias and variance



- Each grey line: one independent sample of 25 observations.
- Restrictive fit: stable across samples, wrong on average.
- Flexible fit: right on average, far off in any one sample.

Two distinct errors, and less of one buys more of the other.

The bias–variance trade-off



- Squared bias, variance, noise.
- Restrictive: bias dominates.
- Flexible: variance dominates.
- The noise is irreducible.
- One point on the curve is best.

Which point, though? The curve is drawn here because the truth is known.

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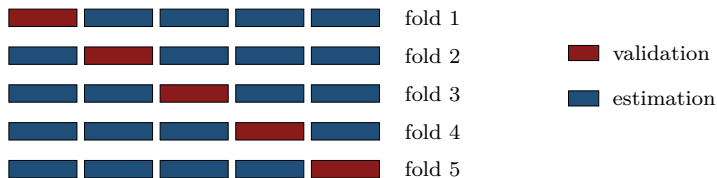
Holding observations back

Training	Validation	Test
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- Out-of-sample error is unobserved. Hide observations and imitate it.
- Training: the coefficients, for one given specification.
- Validation: the choice between specifications, and how flexible.
- Test: one number, once, after every choice is final.

Why a third sample: a sample used to choose is no longer unseen, and its error is optimistic.

Cross-validation



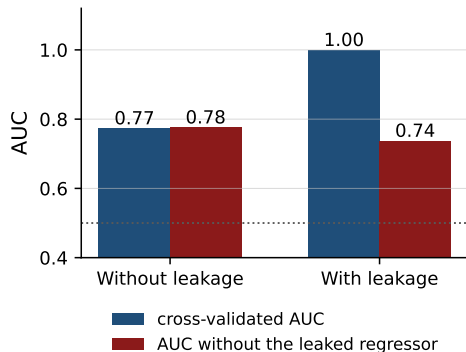
- One validation sample: few observations, and one arbitrary draw.
- Instead, cut the training data into k folds of equal size.
- Each fold is the validation sample once; the other $k - 1$ estimate.
- Average the k errors.

Cross-validation replaces the validation sample, not the test sample. That one is still untouched.

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Leakage



- One regressor built from the outcome.
- Cross-validated fit: almost perfect.
- Drop it: no better than honest.

The held-out fold was never really unseen, and nothing in the procedure said so.

Two kinds of leakage

- Wrong date: a regressor unknown at the moment of the prediction.
- Wrong sample: a quantity computed on data outside the training fold.
- The second, in practice: standardization, imputation, category means.
- Both raise the cross-validated fit and lower the real one.

An unexpectedly good fit is a symptom before it is a result.

Keeping the estimate honest

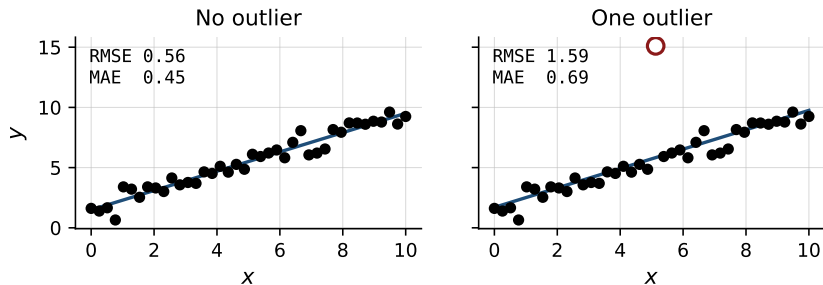
- Split first, transform afterwards. Never the reverse.
- Every transformation fitted on the training fold, applied to the other.
- A regressor enters only if it is known before the outcome.
- The test sample is opened once, at the end.

In code: the transformations go inside the fold, with the estimator.

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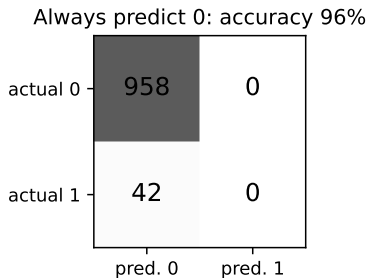
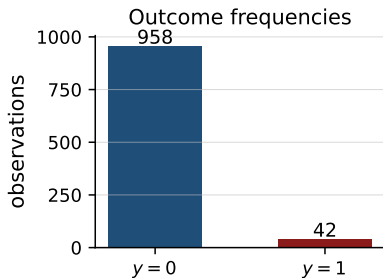
Measuring accuracy: continuous outcomes



- Root mean squared error: large errors count for more.
- Mean absolute error: every unit of error counts the same.
- A few extreme observations move the first, not the second.

Fix the measure before the estimation, by the cost of an error.

Measuring accuracy: binary outcomes



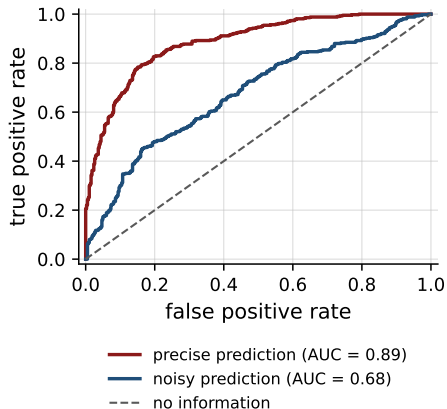
- Rare ones: the share of correct classifications says little.
- Zero everywhere: 96% correct, nothing identified.
- Predicted against actual, as a table: the useful summary.

Thresholds, precision, and recall

- Estimators return a probability. A classification needs a threshold.
- Precision: of those predicted one, the share that are one.
- Recall: of those that are one, the share predicted one.
- A higher threshold: more precision, less recall.

The threshold is a decision about the cost of the two errors, not an estimation problem.

The ROC curve

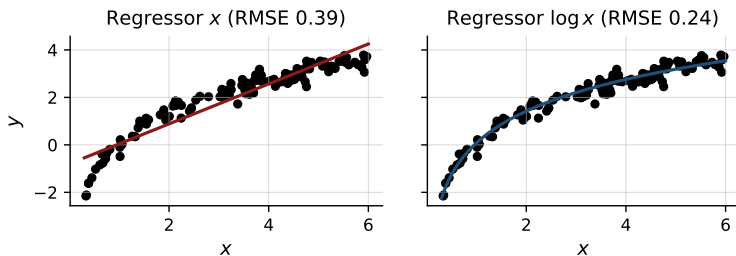


- Both rates, at every threshold.
- Its area: the quality of the ordering.
- No one threshold involved.
- The diagonal: no information.

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The choice of regressors



- Least squares on one regressor, in both panels.
- The outcome: $y = 2 \log x + \varepsilon$.
- x replaced by $\log x$: out-of-sample error down by an order of magnitude.

Same estimator, same data. The transformation supplies the missing knowledge.

What a transformation buys

- Linear models: curvature and interactions only where supplied.
- Trees: both on their own, but learned, and only with enough data.
- A monotone transformation: everything for the first, nothing for a tree.
- Ratios, shares, per capita, real terms: worth supplying to either.

The candidates come from the subject. The folds say which ones earn a place.

Categories, dates, missing values

- Few categories: one indicator each. Many: group the rare ones.
- The mean outcome within a category: leakage, unless computed per fold.
- Dates: year, month, day of the week, as separate regressors.
- Missing: impute, and keep the indicator. Missingness is informative.

A regressor with no content is not free: it is paid for in variance.

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Choosing an estimator

- No estimator dominates. The ranking is a property of the data.
- Least squares first: the benchmark against which flexibility is judged.
- Then one flexible estimator, on the same regressors and the same folds.
- More only when the gain is larger than the variation across folds.

Between two within sampling error of each other, take the simpler one.

What decides the ranking

Restrictive fits win	Flexible fits win
Few observations per regressor	Many observations
Weak signal, much noise	Strong signal
Smooth, additive relations	Thresholds and interactions
Extrapolation is required	Prediction inside the data

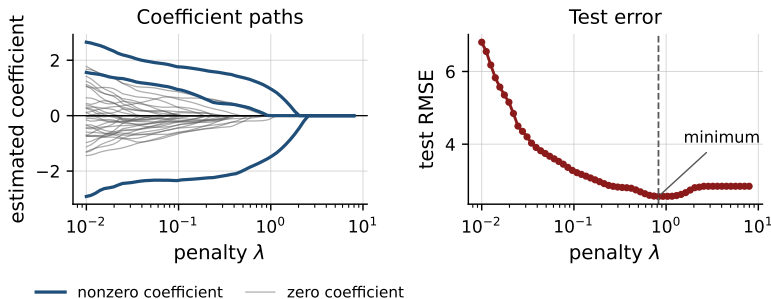
Each row is a conjecture about the data. The cross-validated error settles it.

What each estimator buys

- Least squares: a benchmark, with nothing to choose.
- Ridge and LASSO: a penalty on the coefficients; LASSO also selects.
- Regression tree: thresholds and interactions; on its own, unstable.
- Random forest: the tree's flexibility, most of its variance removed.
- Boosting: usually the most accurate, and the most to choose.

Each has its counterpart for a binary outcome.

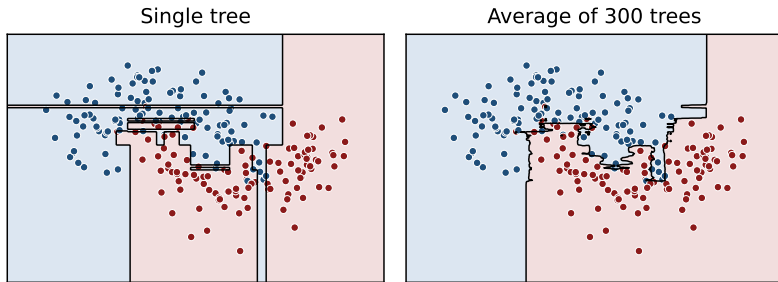
Penalized regression



- 70 observations, 40 regressors, 3 with nonzero coefficients.
- The penalty shrinks the coefficients, and sets the irrelevant to zero.
- Lowest test error at a strictly positive penalty.

The penalty is one more choice, and the folds make it.

Trees and random forests



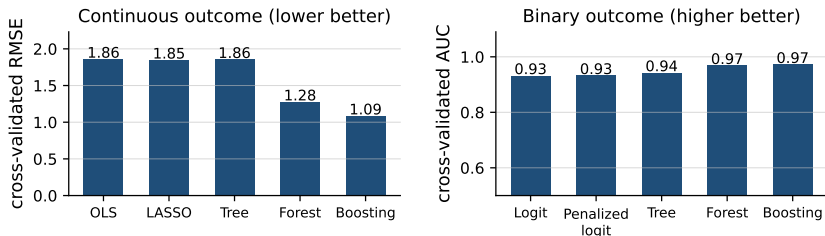
- One tree: the boundary is a feature of the particular sample.
- A forest: many trees on resampled data, averaged.
- The flexibility kept, most of the variance gone.

Boosting

- Trees in sequence, not in parallel.
- Each one fitted to the residuals of the current model.
- Each entering with a small weight, so progress is gradual.
- More parameters than a forest, and it overfits when they are careless.

A forest asks fewer questions of you, and is the better place to start.

Comparing estimators



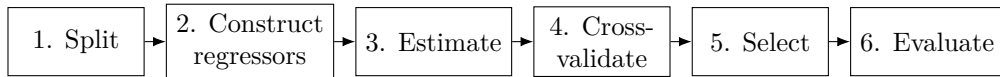
- Same regressors, same folds, for every estimator.
- Flexible estimators win here: curvature and interaction in the data.
- On data close to linear, the ordering reverses.

The table earlier is a guide to where to look. This figure is the answer.

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The workflow



- Was every transformation fitted inside the fold?
- Could any regressor contain the outcome?
- Same regressors and same folds across every specification?

Step 6 happens once. Anything learned there is not a reason to return to step 2.

Summary

- Accuracy concerns observations not used in the estimation.
- In-sample fit does not measure it; bias against variance is the trade-off.
- Cross-validation estimates it. Leakage quietly destroys the estimate.
- Fix the measure of accuracy in advance, by the cost of an error.
- Regressors first, estimator second.

Appendix

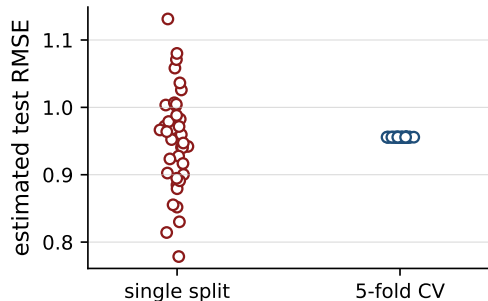
The decomposition of expected prediction error

For a new observation at x_0 , with $y = f(x_0) + \varepsilon$ and $\hat{f}$ estimated on a random sample,

$$\mathbb{E}[(y - \hat{f}(x_0))^2] = (\mathbb{E}[\hat{f}(x_0)] - f(x_0))^2 + \text{Var}(\hat{f}(x_0)) + \text{Var}(\varepsilon).$$

- The expectation is taken over samples, not over observations.
- The first two terms can be traded against one another.
- The third cannot be reduced by any estimator.
- Penalization and averaging lower the second at some cost in the first.

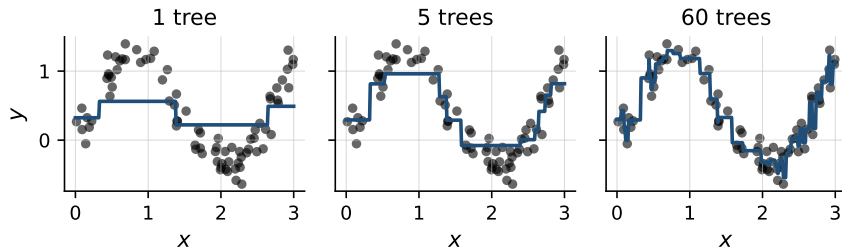
Sampling variability of a single split



- Each point: a new partition.
- One split varies a great deal.
- The average over folds is stable.
- Small samples: it matters most.

With a binary outcome, keep the share of ones equal across folds.

Boosting: successive fits



- A single tree of depth two is a coarse step function.
- Successive trees fitted to the residuals accumulate into an accurate fit.
- Continued far enough, the procedure begins to fit the noise.
- The number of trees is itself chosen by cross-validation.

Further measures for binary outcomes

- When ones are rare, the precision–recall curve is more informative.
- ROC can look favourable when few of the predicted ones are correct.
- The F_1 measure combines precision and recall in one number.
- Log-loss penalizes confident predictions that turn out to be wrong.
- Calibration: of those given probability 0.7, about 70% should be ones.

References

- Mullainathan and Spiess (2017), *Journal of Economic Perspectives* 31(2).
- Athey and Imbens (2019), *Annual Review of Economics* 11, 685–725.
- James, Witten, Hastie and Tibshirani, *Introduction to Statistical Learning*.
- Hastie, Tibshirani and Friedman, *The Elements of Statistical Learning*.